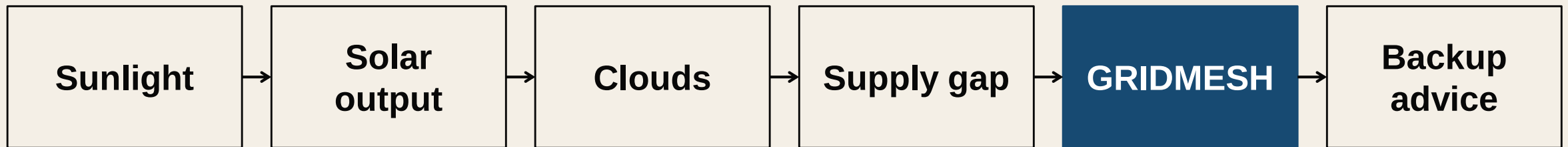


GRIDMESH

**When clouds cut solar,
who keeps the neighbourhood powered?**



GridMesh helps local operators turn uncertain solar forecasts into backup advice within available power and energy limits.

The cloud arrives. The gap doubles.

02/12

Illustration: demand 100 kW. Grid limit 60 kW.

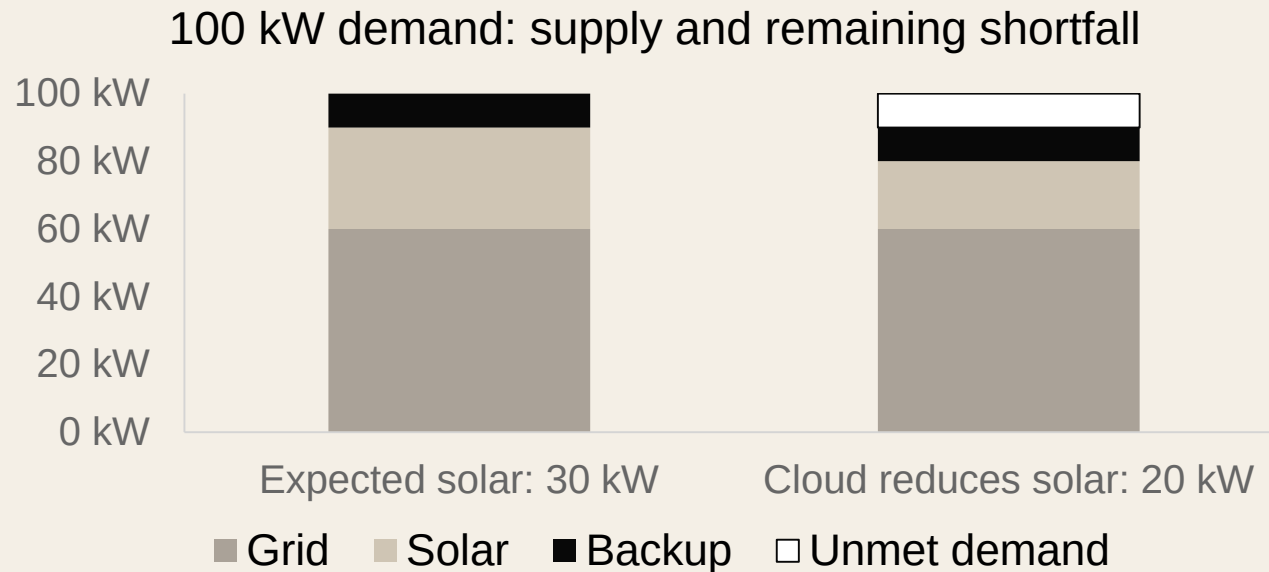
BEFORE CLOUD / EXPECTED GAP

10 kW



AFTER CLOUD / ACTUAL GAP

20 kW



Only 10 kW backup is ready

Solar falls from 30 to 20 kW.
10 kW demand remains unmet.

Backup is limited. Forecasts are uncertain.

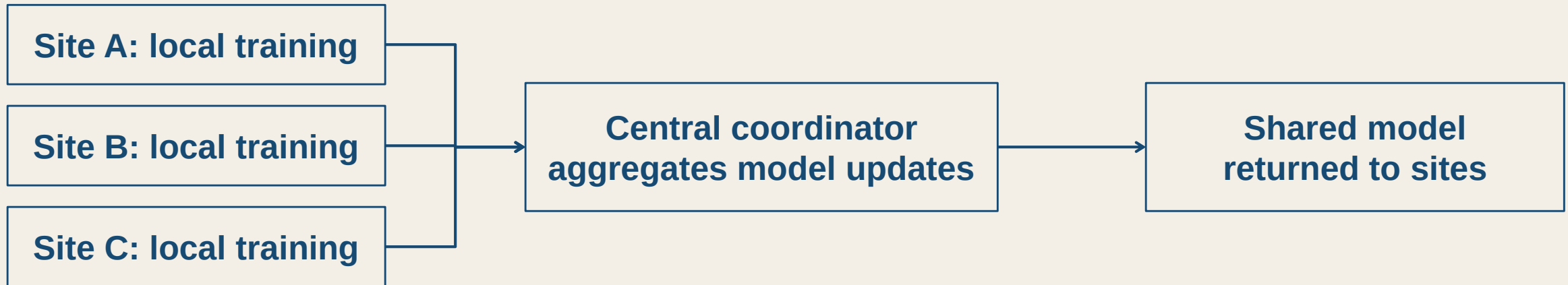
Illustrative arithmetic example - not an experimental result.

GridMesh turns a forecast into a plan

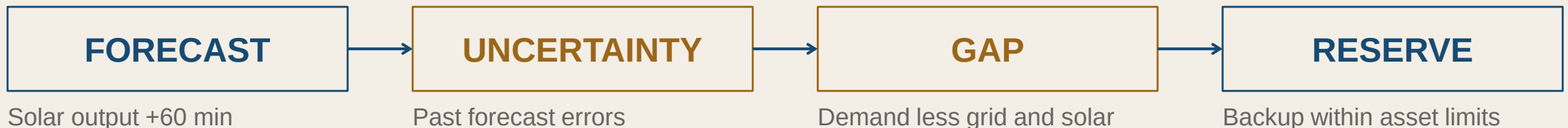
03/12

FEDERATED LEARNING

Separately managed sites learn together while keeping raw histories local. Sites share model updates. FedAvg combines them into a shared model.



The shared forecasts feed one reserve-planning node.

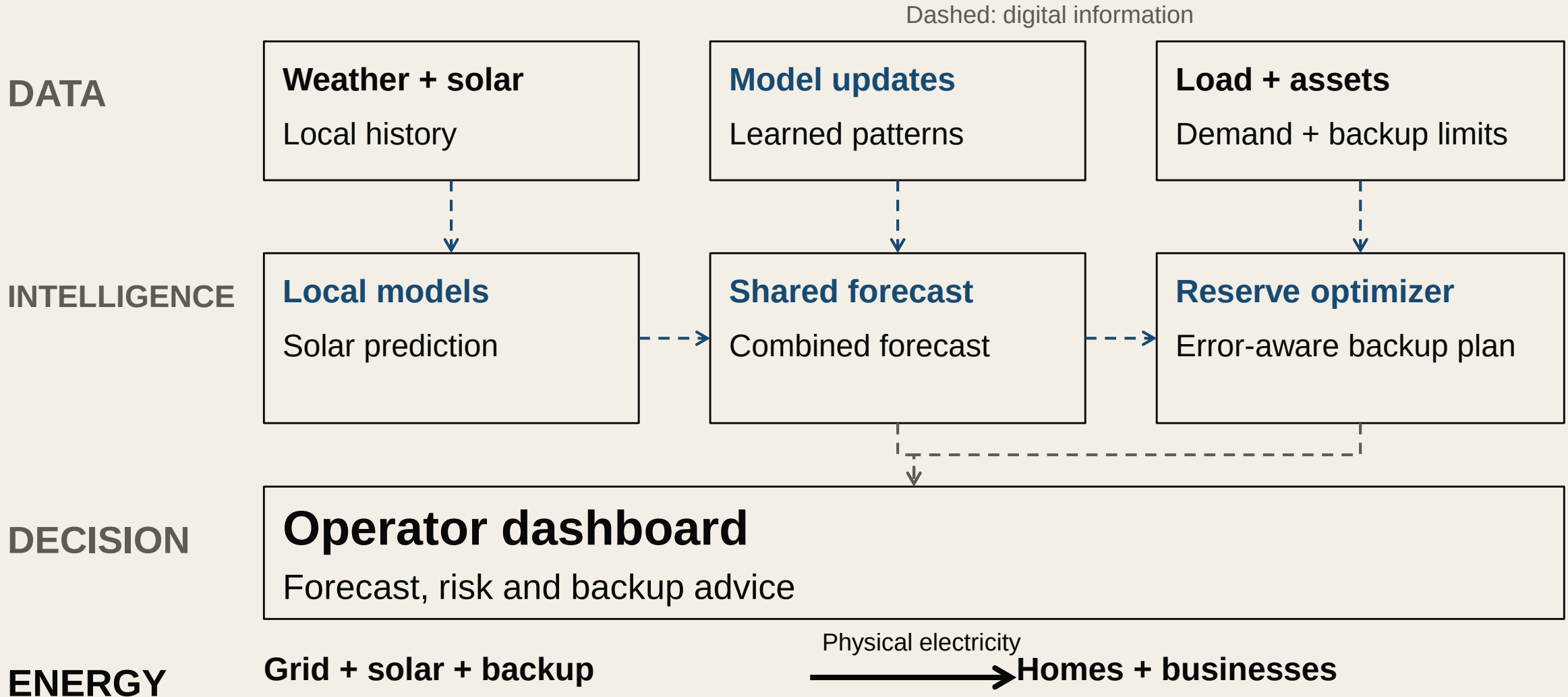


100 virtual sites in one process. No formal privacy guarantee, geographic validation or live grid control.

GridMesh architecture

04/12

Local data becomes a shared forecast and risk-aware backup recommendation.



Advisory prototype: operators and equipment determine the physical response. No live control integration.

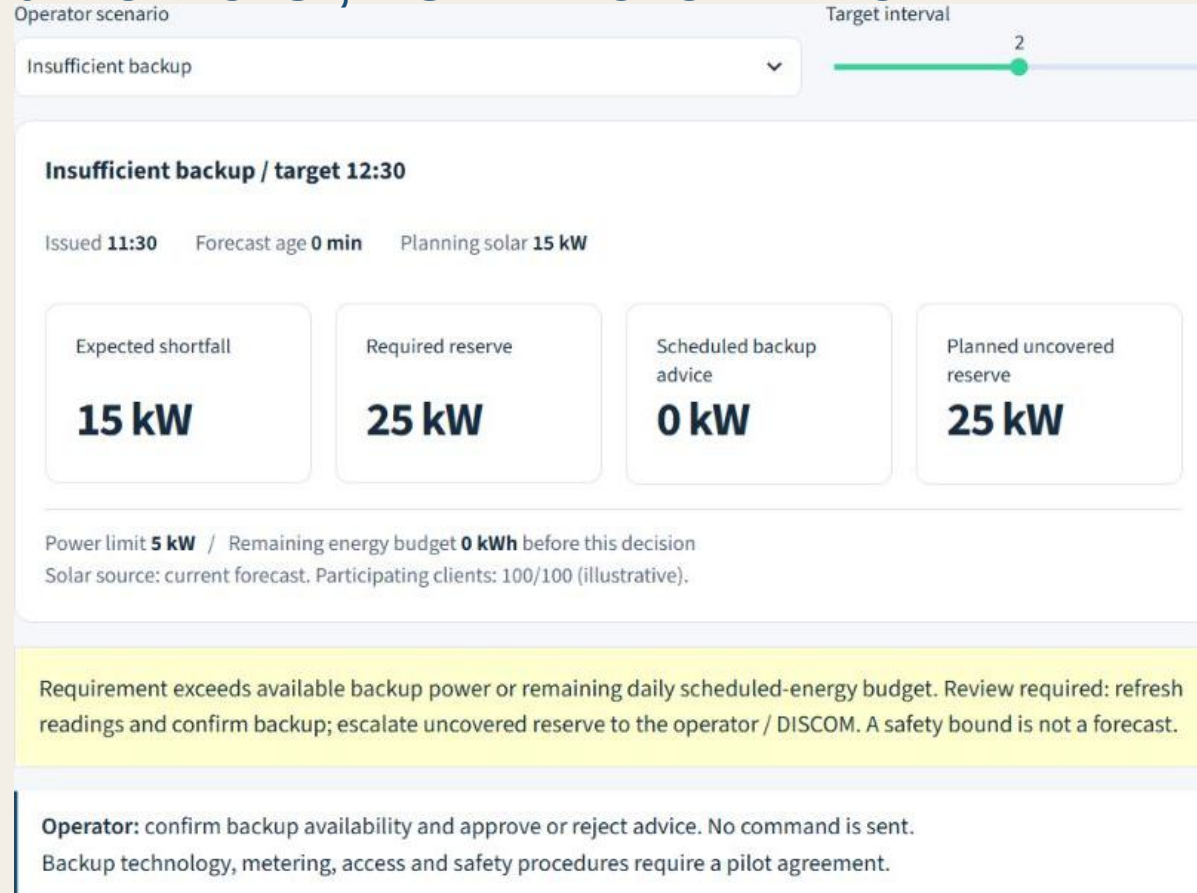
GridMesh recommends. The local operator decides.

05/12

IMPLEMENTED OPERATOR WORKSPACE

Five scenarios · synthetic 100-home assumptions · +60-minute advice

01 FORECAST, RISK + BACKUP ADVICE



02 SUPPLY + RESERVE GRAPHS



**Demand exceeds solar + grid.
Uncovered reserve stays visible.**

03 BACKUP-ENERGY BUDGET



Assumed budget, not battery charge.

Synthetic replay and advice only. No equipment commands or field reliability claim.

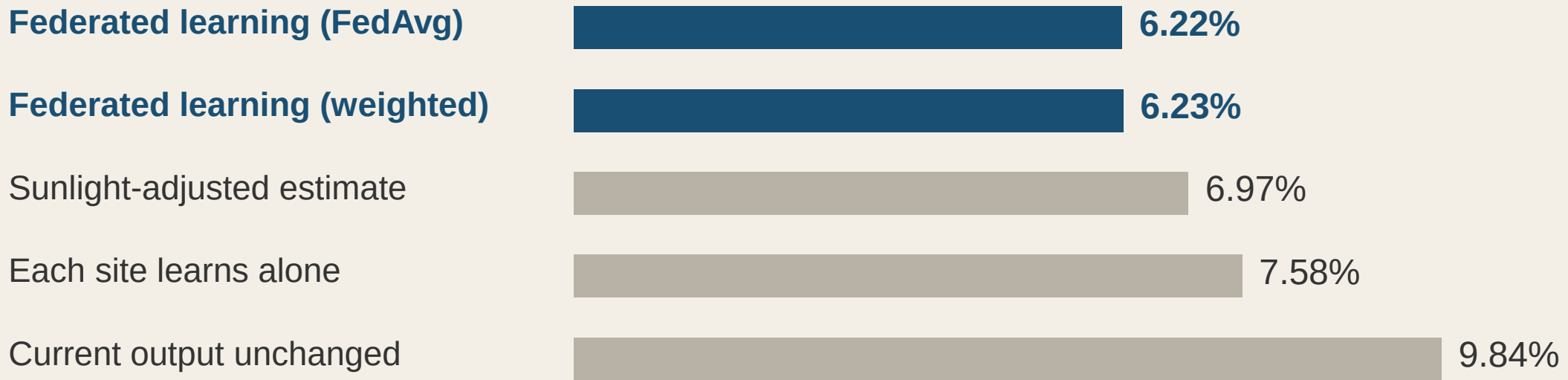
Federated learning gave lower forecast error

06/12

LOWER ERROR IN THIS SIMULATED EXPERIMENT

2024 Bengaluru weather, 100 simulated sites, one-hour forecast

Daytime forecast error (RMSE), averaged across 5 runs. Shorter bars are better.



About 18% lower error than learning at each site alone.

FedAvg: 6.22% error vs local-only: 7.58%.

Weighted averaging performed almost equally (6.23%).

RMSE = forecast error as % of solar capacity. Training budgets differ.
Simulated sites and modelled solar; results do not establish field performance.

Same forecast, smarter reserves

07/12

CURRENT INDIA / CHRONOLOGICAL SIMULATION

96.89% → 99.39% +2.50 percentage points

FULLY SUPPLIED DAYTIME INTERVALS

Approximately 38% less scheduled backup.

Metric	Fixed 20% reserve	GridMesh nominal reserve
Fully supplied intervals	96.89%	99.39%
Unserved energy	0.875 MWh	0.128 MWh
Scheduled backup	27.04 MWh	16.76 MWh

Same reliability-weighted MLP forecasts, causal planner and assets; only the reserve rule changes.

Five-seed chronological simulation. Modelled solar, synthetic demand and assumed backup —not measured uptime.

A reserve that responds to uncertainty

08/12

	FIXED RESERVE	GRIDMESH
Reserve logic	20% of reference demand at each interval	Margin from past forecast errors
Uncertainty response	No adjustment to errors	Larger spread can increase margin
Confidence response	Same percentage rule	Smaller spread can reduce margin
Constraints	Same power + daily energy limits	Same power + daily energy limits

Reserve margin: extra backup for forecast uncertainty.

$$\text{Margin} = \max(0, 1.645\sigma - \mu)$$

μ : mean error. σ : error spread. Error = actual solar – forecast.

Nominal statistical coverage does not guarantee achieved supply availability.

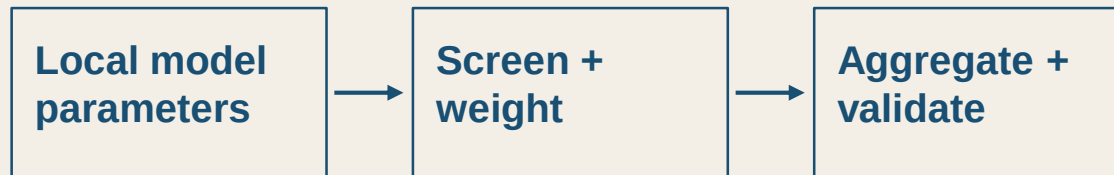
Robust aggregation and communication-efficient FL

09/12

Reliability-aware parameter averaging and selective client uploads

1. RELIABILITY-AWARE AGGREGATION

Stale or degraded client updates can distort the global model.



Validation failure triggers model rollback / fallback.

NO CONSISTENT RMSE ADVANTAGE

Stress tests: stale clients and 50% client dropout.
Reliability weighting remains experimental.

100 virtual clients, 3 paired random seeds.

Single-process simulation. Protocol bytes are modelled, not measured network traffic. Forecast robustness does not establish supply availability.

2. SELECTIVE MODEL TRANSMISSION

Prioritise changed clients; stable clients skip large model uploads.

Normalised protocol bytes: basic FedAvg = 100%

Basic FedAvg



Selective FL



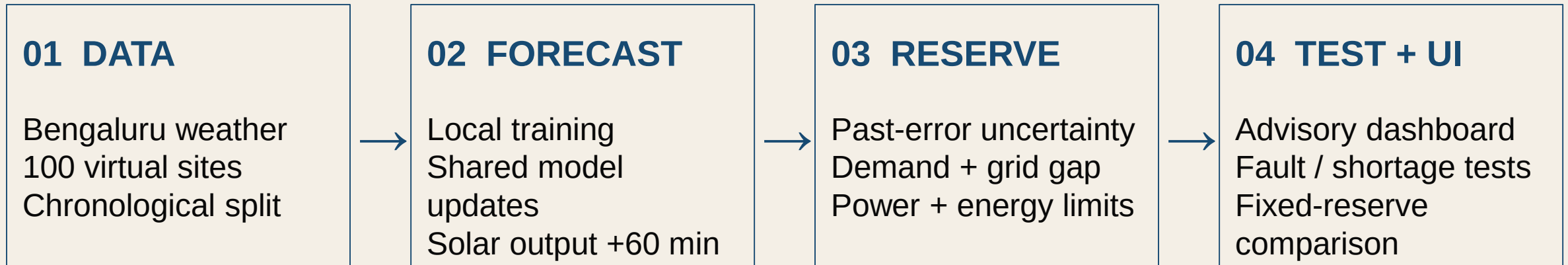
45.55% fewer protocol bytes

Accuracy trade-off: +2.72% relative test RMSE versus basic FedAvg.

Prototype implementation roadmap

10/12

IMPLEMENTED IN THE SIMULATION-BASED PROTOTYPE



Built: forecasts, constrained backup advice and an operator dashboard.

NEXT: FIELD VALIDATION

Measured local data → confirm assets + costs → operator shadow pilot

PROPOSED COMPLEMENT TO SCHNEIDER

Local forecasts + reserve advice for existing operator / DISCOM workflows.

Simulation and advice only. No live control, implemented integration or field deployment.

Can a neighbourhood afford it?

11/12

ILLUSTRATIVE SERVICE BUDGET

₹15,268 / month

Cloud + connectivity	₹3,615.52
Paid operation + support	₹9,060.00
Setup recovery (36 months)	₹601.39
15% contingency	₹1,991.54

≈₹153 / paying home / month

100 payers · before any funding

≈₹76 with 50% co-funding, if secured.

PROPOSED OWNERSHIP + O&M

Stakeholder	Role
Community cooperative / local energy company	Owns the service
Local operator	Reviews forecasts and reserve advice
Technician	Maintains sensors
Software partner	Models, software, security
DISCOM	Approves grid coordination

Planning estimate, not a quote. Funding and willingness to pay remain unconfirmed.

Existing assets assumed. Hardware, retrofit, backup energy and asset-access costs excluded / TBD.

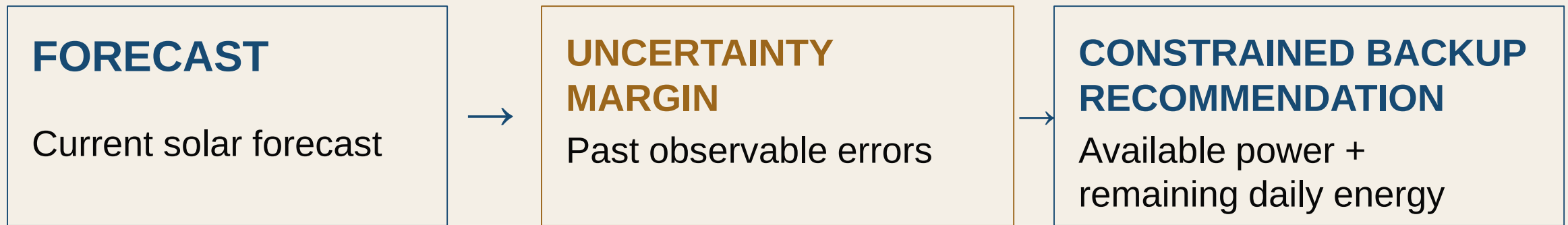
₹15,268 is the rounded covered service budget, not an all-in price or validated affordability.

Reserve optimization: from forecast to decision

12/12

CURRENT INDIA · 100 VIRTUAL SITES · +60 MIN

Chronological 70/15/15 split; single-interval planning.



Primary benchmark assumes known load / grid. New stresses vary both.

8 operating scenarios × 5 seeds: expose asset and forecast limits.

Experimental guard misses its target under several stresses.

TEAM BP

Bhavesh M · Ajay Javali · Pragun kirani · Varshith M gowda

Mentors: Prof. Aswin Kannan · Prof. Shrisha Rao

Simulation only. Field reliability and total affordability remain unvalidated.